

Biweekly Report Two

Reporting Period: 07 June 2026 – 20 June 2026

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Program: M.Tech Smart Mobility and Logistics Systems

Project: Building Foundation Models for Urban Mobility and Smart Cities

Things I learnt in this period are as follows,

1. Learnt about transformer architecture
2. Code a basic GCN on the toy (ESOL) dataset
3. Read the following three research papers :
 1. GCN (ICLR'17) — Kipf & Welling
 2. TTM: Tiny Time Mixers (NeurIPS'24) — IBM
 3. PatchTST (ICLR'23)

Things I plan to learn in next two weeks are as follows,

1. **Establish Baseline:** Implement zero-shot inference using the pre-trained IBM TTM model on the METR-LA test split to establish a baseline MAE (Mean Absolute Error) without spatial awareness.
2. **Build Spatial Branch:** Write the graph processing module to compute the symmetrically normalized adjacency matrix and implement a node-level GCN layer to extract spatial features from the road network.
3. **Parallel Adapter Integration:** Build the first complete prototype of the Parallel Adapter architecture (combining the frozen TTM temporal branch, trainable GCN spatial branch, and a fusion MLP head) and initiate training on METR-LA to see if we can beat the vanilla TTM baseline

4. Continue with the literature review by reading any four papers out of the followings:
 1. **UniST** — State-of-the-art spatio-temporal foundation model baseline.
 2. **STGCN (IJCAI'18)** — First framework combining GCNs with temporal convolutions.
 3. **TSMixer-IBM** — Efficient, MLP-based time-series forecasting architecture.
 4. **DCRNN (ICLR'18)** — Combines diffusion graph convolutions with GRUs.
 5. **Graph WaveNet (IJCAI'19)** — Introduces self-adaptive learned graph adjacency.
 6. **DLinear** — Are Transformers Effective for TS Forecasting? — Proves simple linear layers beat complex Transformers.
 7. **Low Rank Adaptation for LLM (LoRA)** — Parameter-efficient tuning method inspiring our adapter.